$\cancel{P(x_{max}\ \ x_{max}}$

если $\frac{(\theta+\varepsilon)n}{(n+1)} > \theta$; $(\theta+\varepsilon)n > \theta(n+1)$; $\theta(n$

$\varepsilon n > \theta \Rightarrow$ при $n > \frac{\theta}{\varepsilon}$ $P\left(x_{max} < \frac{(\theta+\varepsilon)}{n+1}n\right) = 1 \Rightarrow$

$\Rightarrow P(\tilde{\theta}_3' \geq \theta+\varepsilon) \xrightarrow[n\to\infty]{} 0$

2) $P(\tilde{\theta}_3' \leq \theta-\varepsilon) = P\left(\frac{x_{max}(n+1)}{n} \leq \theta+\varepsilon\right) =$

$= P\left(\underbrace{x_{max} \leq \frac{(\theta-\varepsilon)n}{n+1}}\right) = \cancel{1 - P\left(x_{max} > \frac{(\theta-\varepsilon)n}{n+1}\right)}$

~~выполнено при~~ $\cancel{\frac{(\theta+\varepsilon)n}{n+1} < \theta}$;

$\cancel{(\theta+\varepsilon)n < \theta(n+1)}$

$\cancel{\varepsilon n < \theta \Rightarrow n < \frac{\theta}{\varepsilon} \Rightarrow}$ ~~при~~

$= \left(F\left(\frac{(\theta-\varepsilon)n}{n+1}\right)\right)^n = \left(\frac{(\theta-\varepsilon)n}{\theta(n+1)}\right)^n = \left(\frac{\theta-\varepsilon}{\theta}\right)^n \cdot \left(\frac{n}{n+1}\right)^n =$

$= \left(\frac{\theta-\varepsilon}{\theta}\right)^n \cdot \cancel{\frac{1}{\left(1+\frac{1}{n}\right)^n}} \left(\frac{n+1}{n}\right)^{-n} = \left(\frac{\theta-\varepsilon}{\theta}\right)^n \left(\left(1+\frac{1}{n}\right)^n\right)^{-1} =$

$= \left(\frac{\theta-\varepsilon}{\theta}\right)^n / e = \left\{ \text{т.к. } \left(\frac{\theta-\varepsilon}{\theta}\right) < 1 \Rightarrow \left(\frac{\theta-\varepsilon}{\theta}\right)^n \to 0 \right\} \Rightarrow$

$\Rightarrow$ при $n \to \infty$: $P(\tilde{\theta}_3' \leq \theta-\varepsilon) \to 0$

Итог: оценка состоятельна

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2) $\tilde{\theta}_4 = x_1 + \frac{\sum_{k=2}^{n} x_k}{(n-1)}$

Несмещенность:

$M\tilde{\theta}_4 = \cancel{M\xi}\, M\left(x_1 + \frac{\sum_{k=2}^{n} x_k}{(n-1)}\right) = M\xi + \frac{(n-1)\cancel{M\xi}\, M\xi}{(n-1)} =$

$= 2M\xi$;

$M\xi = \int_0^\theta x \cdot \frac{1}{\theta}\, dx = \frac{x^2}{2\theta}\Big|_0^\theta = \frac{\theta}{2} \Rightarrow M\tilde{\theta}_4 = \frac{2\cdot\theta}{2} = \theta \Rightarrow$

$\Rightarrow$ оценка получилась несмещенная